

Problems in Exponential Distribution:

- 1)

If the number of kilometres that a car can run before its battery wears out is exponentially distributed with an average value of 10,000 km and if the owner desires to take a 5000 km trip, what is the probability that he will be able to complete his trip without having to replace the car battery. Assume that the car has been used for some time. What is the probability, when the distribution is not exponential?
- 2)

If X is exponentially distributed, prove that the probability that X exceeds its expected value is less than 0.5.
- 3)

The amount of time that a watch will run without having to be reset is a RV having an exponential distribution with mean 120 days. Find the probability that such a watch will

 - (i) have to be set in less than 24 days and
 - (ii) not have to be reset in at least 180 days.
- 4)

The daily consumption of milk in excess of 20,000 litres in a town is approximately exponentially distributed with parameter $1/3000$. The town has a daily stock of 35,000L. What is the probability that of 2 days selected at random, the stock is insufficient for both days?
- 5)

The length of the shower on a tropical island during rainy season has an exponential distribution with parameter 2, time being measured in minutes. What is the probability that a shower will last more than 3 min? If a shower has already lasted for 2 min, what is the probability that it will last for at least one more minute?